Sections: B,C & D

Linear Algebra (MT-1004)

Homework#30

Instructor: Muhammad Ali

Regression using QR Factorization

Question 1

A retail store wants to predict monthly sales (in thousands) based on two factors: the number of employees and the number of products displayed. The dataset is as follows:

Number of Employees	Products Displayed	Sales (Y in thousands)
2	3	30
3	5	7
5	8	14
7	13	22
9	21	16

- Write the design matrix for given data
- Compute the coefficients b (Bias) and w_1 (Weights) using the idea of QR factorization
- Perform a multivariate linear regression model on the provided dataset
- Predict the sales when there are 16 employees and 6 products displayed

Hint: As you know that $w = (A^T A)^{-1} A^T Y$. If we use QR factorization w can be written as:

$$w = R^{-1} Q^T$$